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Comprehension of the Shakespeare authorship question through deep impostors approach

Zeev Volkovich [1,*], **Renata Avros** [1]

[1]Software Engineering, Braude College, Karmiel 2161002, Israel

*Corresponding author. Software Engineering, Braude College, Karmiel 2161002, Israel. E-mail: vlvolkov@braude.ac.il

Abstract

This article investigates the authorship question surrounding William Shakespeare's works using a novel approach called the 'Deep Impostor' methodology. The approach uses a set of known impostor texts to analyze the origin of a target text collection. Both the target texts and impostors are divided into an equal number of word segments. A deep neural network, either a Convolutional Neural Network (*CNN*) or a pre-trained *BERT* transformer, is then trained and fine-tuned to differentiate between impostor segments. Once assigned, each target text is transformed into a numerical signal by averaging its segment assignments. The Dynamic Time Warping distance is afterward evaluated between these signals to measure their similarity. The Isolation Forest algorithm identifies outliers within the target text collection for each impostor pair by assigning appropriate scores to each tested text. In the summarizing step, the tested creations are clustered into two groups. In the case of a *CNN*-based model, the first resulting cluster contains fifteen creations. General evaluations lead to the conclusion to suggest that these are not authored by Shakespeare. The remaining documents are classified as authentic Shakespearean works.

Keywords: the Shakespeare authorship question; deep impostors approach.

1. Introduction

In the timeframe that followed William Shakespeare's demise, scholars have engaged in a captivating debate over the actual authorship of his works. The question of Shakespeare's authorship has captivated researchers and scholars for a long time, generating a lot of studies and discussions. The film 'Anonymous' released in 2011, written by John Orloff and directed by Roland Emmerich, has significantly contributed to the heightened intensity of the debate surrounding the actual authorship of Shakespeare's works. By presenting a thought-provoking narrative that explores the possibility of alternative authors, 'Anonymous' has sparked renewed interest and added a layer of excitement to the ongoing discourse on Shakespeare's authorship. The film has successfully brought the topic to a broader audience, including those with limited interest or awareness of the issue. By positing an alternative narrative and exploring the possibility of Edward de Vere, the 17th Earl of Oxford, as the potential actual creator of Shakespeare's plays, 'Anonymous' has made this debate more accessible and popular among a broader range of viewers and has played a vital role in reigniting attention and sparking renewed interest in this matter.

The 'Shakespeare authorship question' has captured significant attention in literature and the Internet, leading to many books, articles, and online discussions dedicated to dissecting and analyzing the matter. As noticed in the famous book '*Who Wrote That?*: Authorship Controversies from Moses to Sholokhov' by Ostrowski (2020):

> *After all, "Shakespeare wrote Shakespeare" has become one of the pillars of most people's worldview, akin to the earth is round and 2 + 2 = 4. Anyone who challenges what from their point of view is obviously true must be a crank or charlatan or someone who is being argumentative for argument's sake. Yet, is there another way to account for the persistence of the skeptics?*

The topic of Shakespeare's authorship has garnered extensive attention in both print and online mediums, with a multitude of monographs, articles, posts, and discussions dedicated to exploring this subject. The vastness and complexity of this topic make it impossible to delve into every detail within the scope of this article. At this juncture, we present a selection of publications that provide a glimpse into the wide range of materials available, showcasing the extent and complexity of the ongoing discourse surrounding this topic.

In addition to Donald Ostrowski's book (Ostrowski 2020), other notable manuscripts delve into the question of Shakespeare's authorship. Here are just a few more examples:

- 'Shakespeare Beyond Doubt? Evidence, Argument, Controversy', edited by Wells et al. (1997). This diverse collection features essays from prominent scholars and experts who examine the evidence and arguments surrounding the authorship question, address the doubts and controversies associated with Shakespeare's works.
- 'Contested Will: Who Wrote Shakespeare?' by Shapiro (2011). In this book, Shapiro delves into the authorship question and provides an in-depth exploration of the theories, controversies, and debates surrounding the true identity of the author behind Shakespeare's plays and poems.
- 'Who Wrote Shakespeare?' by Michell (1999). Michell's work is a thought-provoking exploration of various alternative theories regarding the authorship of Shakespeare's works. Challenging the traditional attribution, he presents arguments and evidence supporting alternative candidates, raising intriguing questions about the Shakespearean authorship debate.
- 'The Shakespeare Wars: Clashing Scholars, Public Fiascoes, Palace Coups' by Rosenbaum (2011). While not solely focused on authorship, this book investigates the controversies and clashes between scholars regarding the interpretation and authenticity of Shakespeare's works, including discussions on the authorship question.
- 'The Mysterious William Shakespeare: The Myth & the Reality' by Ogburn (1992). This book challenges the traditional attribution of Shakespeare's works and presents alternative theories and candidates for the authorship, delving into historical and literary evidence.
- 'Shakespeare's Unorthodox Biography: New Evidence of an Authorship Problem' by Price (2001). Diana Price examines the authorship question by analyzing historical documents and contemporary records, raising doubts about the traditional attribution of Shakespeare's works.
- James and Rubinstein's (2017) 'The Truth Will Out: Unmasking the Real Shakespeare' examines the authorship question. In this work, the authors explore the controversy surrounding the author's true identity behind Shakespeare's works. They present arguments, evidence, and alternative theories, aiming to shed light on the potential candidates and challenge the traditional attribution of Shakespeare as the sole author.
- Vickers's (2004) book posits that William Shakespeare did not solely author all the plays attributed to him. Vickers explores the collaborative practices of Elizabethan theatre and employs stylometric analysis to identify instances of co-authorship.

Note that the books 'Who Wrote Shakespeare?' by Michell (1999), 'The Mysterious William Shakespeare: The Myth & the Reality' by Ogburn (1992), and 'The Truth Will Out: Unmasking the Real Shakespeare' by James and Rubinstein (2017) specifically address the authorship question from an Anti-Stratfordian perspective challenging the traditional attribution of Shakespeare's works and presents alternative theories and candidates for the authorship, exploring into historical and literary evidence.

Anti-Stratfordian researchers cast doubt on the authorship of specific works attributed to Shakespeare, presenting alternative perspectives. According to their viewpoints, these compositions may have been mistakenly credited to Shakespeare or resulted from collaborative efforts involving multiple writers. The most radical stance among these researchers posits that the plays traditionally associated with Shakespeare were actually a collective endeavor rather than the work of a single author.

Using stylometric methods to investigate the Shakespeare authorship question has gained considerable interest. Stylometry involves analyzing quantifiable characteristics of writing styles, such as:

- Word frequency analysis: Examining the frequencies of specific words or word categories across different plays to identify patterns or distinctive usage.
- Vocabulary richness: Assessing the diversity and richness of vocabulary employed in the plays, such as measuring the number of unique words or lexical variety.
- Stylistic markers: Analyzing the presence of specific linguistic features or stylistic markers that may be characteristic of a particular author or group of authors.
- *N*-gram analysis: Examining sequences of words (*N*-grams) to identify recurring patterns or linguistic preferences within the plays.

This methodology was initially pioneered by Thomas Corwin Mendenhall, who compared the works of Shakespeare and Bacon (Mendenhall 1901) and highlighted a notable difference in word length frequency. The results obtained here are skewed by the comparison of verse (e.g. Shakespeare, Marlowe) to prose (e.g. Bacon) (Williams 1975).

This finding suggests that Bacon is unlikely to have authored Shakespeare's works. However, using the same criteria, the word length distribution of Marlowe's works aligns closely with Shakespeare's, indicating the possibility of co-authorship.

Another study by Merriam (1993) examines linguistic and stylometric evidence to assess whether Marlowe may have contributed to the play's writing. Matthews and Merriam (1993) propose using multi-layer perception neural networks for analysis. Aljumily (2015) presents an innovative approach to examining the Shakespeare authorship question. The study incorporates a combination of modern statistical approaches, including hierarchical and non-hierarchical linear and non-linear clustering methods. Furthermore, the study includes an extensive literature review, providing a comprehensive overview of existing research and scholarly discussions surrounding the authorship question.

A comprehensive overview and systematic presentation of quantitative methods for analyzing text authorship can be found in Grieve (2005).

Juola et al.'s (2008) article is a valuable resource for deeper exploration into authorship attribution. This broad review examines numerous quantitative methods, including stylometry and machine learning, employed to determine an author's identity based on writing style.

Koppel and Seidman (2013, 2018) have analyzed the 'Shakespeare authorship question' through the Impostors method introduced by Koppel and Winter (2014). This method, further discussed in Section 2, found numerous fruitful applications. The central beneficial concept entails sequentially comparing the documents in question with randomly selected external texts (impostors) subsets. This comparison process involves arbitrary sampled features within the Vector Space Model framework.

The method has demonstrated its suitability and efficiency, continuously evolving and adapting. However, it does have certain inherent limitations primarily related to its structural design within the mentioned model. One notable weakness is its inability to capture the nuances of the writing style's short patterns, which often significantly define the author's innate writing fashion (e.g. expressions like 'The devil is in the details').

In our comprehension, short patterns refer to relatively brief text intervals comparable in length to tweets. Given the limited number of words in these intervals, traditional word frequency analysis becomes unreliable. Furthermore, this type of analysis overlooks word dependencies, focusing solely on individual word occurrences and emphasizing one-dimensional word distributions, ignoring mutual common distributions crucial for understanding the stylistic connections between terms. A similar challenge arises in the modern age when analyzing short messages such as malicious bots, SMS messages, and Facebook status updates. Alternative approaches have been proposed in Volkovich (2020) and Avros and Volkovich (2022) to address this issue, particularly in studying Arabic Medieval Documents and examining Al-Ghazali's suspected authorship. Such an approach provides insights into examining classical literary works from a different perspective, specifically focusing on shorter texts' distinct features and characteristics. This alternative method facilitates a deeper investigation into the intricate dynamics and subtle nuances of shorter patterns, leading to novel visions into their significance and relevance in the broader domain of literary analysis.

A specific methodology proposed in this article is implemented, generally speaking, using the primary approach described in these papers. First, the tested and the impostors' texts are divided into sequential batches of equal word sizes. A deep neural network is then trained to discern between these batches of the impostors. Subsequently, the trained network is utilized to assign the batches of the tested documents to each respective impostor, generating a real-valued signal through appropriate averaging of the binary assigned batches into chunks of the tested documents. Deep learning text-to-signal conversion methods examined and compared here rely on the 'sentiment analysis' approach for chunks based on a Convolutional Neural Network (*CNN*) or a pre-trained *BERT* transformer. The Dynamic Time Warping (*DTW*) distance is computed between these signals, and an appropriate Isolated Forest algorithm is employed to identify any outliers within the tested documents for each impostor pair. Finally, the results are summarized by clustering the obtained Isolated Forest scores at each step.

In our approach, the concept of writing style is linked to distribution in a multidimensional linear space, considering, in some approximation, long but not too long consecutive pieces of text. Accordingly, a text authored by a particular writer can be approximately represented as an independently and identically distributed *(i.i.d.)* sample drawn from a suitable distribution. In this context, the works attributed to Shakespeare, considered written in his unique style, form the source class, while other texts deviating from this style are viewed as anomalies.

The deep impostor method employs projections onto manifolds that impostors create through the successive network or a transformer training to identify deviations and inconsistencies in writing style. These impostors serve as reference points or models of writing styles diverging from Shakespeare's. By training a classifier to differentiate between the impostor styles, our approach aims to identify and highlight the style inconsistencies within the analyzed texts.

The remaining sections of the paper are structured as follows: Section 2 introduces several interconnected concepts, such as the authorship recognition problem, the classical impostor's method, and vector space models that are relevant to the research. Section 3 concentrates on presenting the proposed approach in detail. We thoroughly discuss the model's rationale, explaining its fundamental components step by step. This section follows a coherent structure highlighting the key stages, including training a network or a *BERT*-based classifier, converting documents into signals, applying the *DTW* distance for document comparison, and clustering anomaly scores. Section 4 is dedicated to presenting the numerical results of the experiments. Finally, Section 5 concludes the paper and summarizes the main findings and contributions.

2. Preliminaries

An author's personal writing style somewhat conveys a writer's general perception of being an embodiment of the all-purpose writing process, composed of related stages known as pre-writing, drafting, writing, sharing and responding, revising and editing, and publishing (Alred, Brusaw, and Oliu 2009).

An individual style identification suggests recognizing the style's unique prototypes. In the context of authorship attribution, a common scenario arises where the formal classification perspective aligns with a one-class classification task. This task involves two sets: the target one, which comprises documents authored by a specific writer, and the non-target one, which consists of documents from other authors. The underlying assumption is that the non-target class documents possess distinct configurations that place them apart from the target class. Treating authorship attribution as a one-class classification problem focuses on identifying the distinguishing features and patterns specific to the target class while discerning deviations present in the non-target class.

Two-class style recognition aims to develop a classifier that can accurately differentiate between the target and alternative styles. This often involves training a machine learning model on a labeled dataset of samples from both classes. The model learns to capture each style's unique features and patterns and makes predictions on unseen data to classify them into the respective style classes. Several surveys of various methods applied in this field are given, for example, in Stamatatos (2016), Kestemont et al. (2016), and Amelin et al. (2018).

The author's verification problem can also be tackled from two distinct perspectives: intrinsic and extrinsic. Intrinsic methods focus on analyzing a single class of texts, typically those attributed to a specific author. These methods operate within the one-class classification task's framework and aim to estimate the similarity between the given set of texts (the target class) and a tested set of documents. On the other hand, extrinsic techniques take a different approach by introducing a non-target class or a pool of external documents. These external documents are typically from different authors or sources and are a contrasting set to the target class. Extrinsic methods aim to adapt the verification task into a binary classification problem to differentiate between the target class and the external documents.

2.1 Impostors' method

A notable and widely recognized extrinsic verification scheme is the Impostors' method, introduced by Koppel and Winter (2014), which has been, up to now, one of the most highly effective approaches in the authorship recognition area. This technique leverages a pool of external texts called impostors to create a comparative framework for author verification. By contrasting the tested documents with the impostor documents, the Impostors' method aims to determine the likelihood of the tested documents being authored by the target writer.

Seidman (2013) introduced a modified version named General Impostors, which handles multiple available documents by the applicant author and is based on the Many-Candidate method suggested by Koppel, Schler, and Argamon (2011). This modified approach won several *PAN* competitions.

Khonji and Iraqi (2014) settle cases appearing once at least one impostor text is acquired more related to the doubtful text than the candidate author. Gutierrez et al. (2015) use a homotopy-based classification to aggregate iterations. Potha and Stamatatos (2017) suggest that motivation by the General Impostors method and the extrinsic verification approach significantly improves state-of-the-art performance.

The source process of Koppel and Winter (2014) constructs a classifier by random projection in each step using a random attributes subgroup and a random set of external documents. The texts are recognized as created in the same style if their similarity is more significant than one calculated across all iterations with the external documents (impostors) in most steps. Here, $Sim(X, Y)$ is a similarity measure between given

Algorithm 1 Authorship Verification Algorithm

Input:
X, Y: A pair of documents under consideration.
S: A set of impostors.
k: The number of iterations.
r: A selection rate.
n: The number of randomly chosen impostors in each iteration.
Δ^*: An authorship significance threshold.
Output: TRUE if X and Y are written by the same author, otherwise FALSE.
$Score \leftarrow 0$
for $j = 1$ to k **do**
  Randomly choose a fraction r of features from the full document's feature collection.
  Randomly choose n impostors $Im_i, i = 1, \ldots, n$ from S.
  for $i = 1$ to n **do**
    if $Sim(X,Y) \times Sim(Y,X) > Sim(X, Im_i) \times Sim(Y, Im_i)$ **then**
      $Score \leftarrow Score + \frac{1}{k}$
    end if
  end for
end for
if $Score > \Delta^*$ **then**
  return TRUE
else
  return FALSE
end if

texts X and Y. Commonly, the Impostors' method uses similarity measures rested upon characteristics obtained within the general *Vector Space Model* proposing vector representations (VR) of a text, at a rough guess, as frequencies or normalized frequencies of the terms, which could be words, N-grams of words, or characters. In Koppel and Winter (2014), the cosine similarity measure is mentioned together with the minimum-maximum measure defined as follows:

$$\text{minmax}(D_1, D_2) = \frac{\sum_{i=1}^{m} \min(VR(D_1)_i, VR(D_2)_i)}{\sum_{i=1}^{m} \max(VR(D_1)_i, VR(D_2)_i)},$$

where $VR(D_j)_i$ is i-th coordinate of a representation of $D_j, j = 1, 2$, and m is the total number of features. In Khonji and Iraqi (2014), this distance and several other distances, for example, a distance metric called 'common N-grams', are used. Moreover, as noticed, the minimax measure was initially introduced in geobotanic by Růžička (1958), reevaluated in Schubert and Telcs (2014), and reminded in a known review (Cha 2007).

The cited publications serve just as a partial list of works that showcase the extensive applications of the method. However, it is essential to acknowledge that the approach also possesses inherent limitations, which stem from the disadvantages associated with the *Vector Space Model* methodology:

- Inability to capture fine-grained details: The *Vector Space Model* methodology, and consequently the approach based on it, may struggle to capture subtle nuances and intricate details within the texts because the model focuses solely on individual word distributions while disregarding term relationships and ignoring the semantic content of the text. It may not effectively account for the writing style's rich complexity and specific linguistic characteristics.
- Dependency on feature selection: The approach's effectiveness depends on selecting appropriate features or attributes for representing the texts. Choosing relevant and informative features is crucial to accurately capturing the writing style and distinguishing between authors.
- Lack of contextual understanding: The *Vector Space Model* methodology treats each text document as an independent entity without considering the context or sequence of the words. This limitation hinders the ability to capture contextual dependencies and the overall flow of the writing, which can be essential in authorship attribution tasks.

2.2 Language deep learning-based models

'Language deep models' are extremely powerful tools in natural language processing (*NLP*) and machine learning. These models leverage deep learning techniques to decode the complexities of human language, enabling them to understand and process it with exceptional accuracy and proficiency.

As previously stated, a method akin to the *Vector Space Model* does not account for the semantic information tied to word sequencing, as it ignores the order of words and their co-occurrences. The conventional Markov N-gram approach considers sequential word organization but generates sparse VR. On the other hand, deep learning-based embedding techniques employ innovative strategies that produce dense VR for individual words while preserving their semantic relationships. These methods, mutually known as '*word embedding*', map words from a predefined vocabulary into a vector space, allowing for effective language modeling.

Like several modern techniques for short-text classification, our method also utilizes a dense word embedding to represent words as real-valued vectors. This sets it apart from the word or *N*-gram substantiated text representations discussed earlier, where each term is associated not with a single dimension but with an entire dense vector. In this way, a text is given as a matrix with semantically skilled columns corresponding to terms lying in a dictionary.

Generally, word embedding is a process employed in *NLP* to depict words as numerical vectors. These vectors encapsulate semantic meanings and associations among words derived from their contextual usage. The fundamental concept behind word embeddings is to translate words from a high-dimensional sparse vocabulary space into a continuous lower-dimensional vector space, allowing words with akin meanings to be positioned closer to one another.

Popular off-the-shelf embedding models can be mentioned: *Word2Vec* (Mikolov 2013), *GloVe* (Pennington, Socher, and Manning 2014), and *FastText* (Joulin 2016). From this perspective, *ELMo* ('Deep contextualized word representations') (Peters et al. 2018) and transformer technology like *BERT* (see, e.g. Devlin et al. 2018), can be reminded. Our research employs the widely known *Word2Vec* technique in the '*skip-gram*' mode. This variant of *Word2Vec* is a powerful method for generating word embeddings and is widely recognized in the field. In the skip-gram approach, the model is designed to predict the context words given a target word, learning from co-occurrence patterns in a vast corpus. As a result, *Word2Vec* effectively captures relationships and similarities between words, making it a valuable tool for various *NLP* tasks, such as word similarity analysis, document classification, and language generation.

Fine-tuning *Word2Vec* entails updating pre-trained word embeddings by training them on a specific task or dataset corresponding to the target domain. This approach allows the embedding to capture better the subtleties, nuances, and peculiarities of the task at hand. To achieve this, we initialize the *Word2Vec* model with pre-trained embeddings and then continue training it on a task-specific dataset, including the tested collection and the current impostors' data. This fine-tuning procedure is carried out at each step of our application.

BERT, short for Bidirectional Encoder Representations from Transformers, marks a pivotal advancement in the realm of *NLP*. This innovative model has substantially enhanced computers' capacity to grasp the intricacies of human language, enabling them to understand meaning and context better. Unlike traditional models that process text in a left-to-right or combined manner, *BERT* considers the entire sentence simultaneously, leading to superior performance over unidirectional models. *BERT* utilizes the Transformer architecture, an advanced attention mechanism, to learn contextual relationships between words (or sub-words) within a text.

The standard Transformer comprises two components: an encoder that processes the input text and a decoder that generates predictions for specific tasks. However, since *BERT* aims to create a language model, it employs only the encoder component. In contrast to directional language models that process text sequentially, the Transformer encoder adopts a non-directional approach, analyzing the entire sequence of words simultaneously. This enables the model to capture contextual information for each word by considering its entire context within the sequence.

Training language models presents challenges, particularly in defining a prediction goal. Many models predict the next word in a sequence, a directional approach that limits context learning. *BERT* addresses this challenge by employing two training strategies: Masked Language Modeling (*MLM*) and Next Sentence Prediction (*NSP*). Due to the substantial number of parameters involved in *BERT* modeling, training either a large or small *BERT* model necessitates vast amounts of text, such as the WIKIPEDIA corpus, and significant computational resources.

Pretrained *BERT* can be adapted for various language tasks by adding a small layer to the core model. Classification tasks like sentiment analysis involve passing the model through this fine-tuning step.

3. Proposed model

As previously stated, the proposed model is designed within a comprehensive framework inspired by the Impostors' method. The proposed approach aims to address and mitigate the flaws and limitations discussed earlier in the previous model. By leveraging the principles and techniques of the Impostors' method, the suggested model seeks to overcome the inherent weaknesses and improve upon the performance and reliability of the previous model.

Let us consider a scenario with a vocabulary *Dic* and an embedding function *W* to map this vocabulary to a Euclidean space R^N:

$$W : Dic \rightarrow R^N.$$

In the case of a specific author, their unique writing style can be deemed within our model by a probability distribution $\boldsymbol{P}$ on R^N, which captures the ordered occurrences of words from *Dic* in their writings. While the occurrences of individual words may be influenced by each other, it is reasonable to assume that sufficiently long text segments (such as those containing about 100 words) exhibit a nearly unbiased joint distribution. Consequently, these segments (batches) can be considered as *i.i.d.* sequences. Hence, a given text D can be viewed as such a sequence:

$$D = \{d_1, \ldots, d_t\} \quad (1)$$

of text batches with the same size $L(t = [|D|/L])$ having the marginal mutual distribution, P_L. The last segment containing fewer than L words is omitted.

Within the framework of one-class classification, documents that deviate in their style from the writing style of the source are regarded as outliers. The primary objective is to accurately locate these outliers in space through projections onto suitable manifolds or subspaces and distinguish them from the target class, which comprises authentic creations in the desired style.

Let us consider collection $\boldsymbol{D} = \{D_1, \ldots, D_n\}$ allegedly containing documents written most probably in large part by a suggested author and forgeries, that is, texts prescribed to the author.

Let $\boldsymbol{Z} = \{Z_1, \ldots, Z_m\}$ be an impostors' set of collections. These corpora are used as reference points in recognizing authentic and non-authentic texts. Each impostor document passes the same procedure and, after a pre-processing stage, is represented in Equation (1) fashion. It is also suggested that the documents considered be abridged to words from a given dictionary *Dic* and transformed by applying a mapping $\boldsymbol{W}$ into sequences of vectors in R^N.

3.1 Converting documents into signals in impostors' manifolds

We denote S as a supervised learning algorithm designed to distinguish between two datasets residing in an N-dimensional Euclidean space. For our purposes, we train the algorithm S on pairs of impostor text collections (Z_i, Z_j). This training allows S to create a separator, called $S_{i,j}$, which may tell apart batches of texts from Z_i and Z_j. In the subsequent step, the trained algorithm $S_{i,j}$ assigns labels (0 and 1) to batches of test texts based on their similarity to Z_i or Z_j, as illustrated in Fig. 1.

In this scenario, the points in the bright area represent batches associated with Z_i, while the points in the dark area represent batches associated with Z_j. The separating surface divides the test collection by assigning the appropriate hues to its points. As a result, each text from the test collection is transformed into a binary sequence according to Equation (1). To obtain a signal representation of a text, the sequence is divided into consecutive chunks, each containing the same fixed number of batches, denoted by k. The signal is then computed as the average of the batch scores within each chunk, taking values from the set $\{0, 1/k, 2/k, \ldots, 1\}$. Examples of document signal representations are shown in Fig. 2.

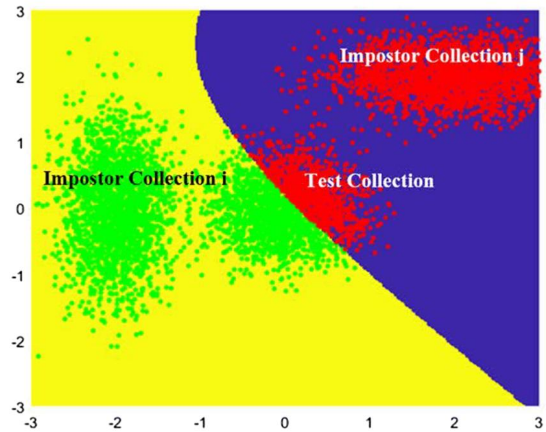

Figure 1. Method illustration.

The general algorithm can be outlined as follows:

Algorithm 2 Transforming testing documents into actionable signals

Input:
  D—a tested collection of documents.
  $Z = \{Z_1, \ldots, Z_m\}$ - impostors' collections.
  S—a supervising separating algorithm.
  L—the batches length.
  k—the number of batches in a chunk.
Output: Signal representations of texts under consideration.
Procedure:
Divide all texts under consideration $d \in D$ and impostors texts $z \in (Z_i, Z_j)$ into sequential batches containing L terms.
Train S on the batches of the impostors' texts to obtain a separator $S_{i,j}$ that distinctly classifies the batches into different impostor collections.
for $d \in D$ **do**
  Initialize $B(d) = \emptyset$.
  for each batch b of d **do**
    Compute the score of batch b using $S_{i,j}$.
    Append the score to $B(d)$.
  end for
  Calculate a signal value, $Sig(d)$, for the document d chunk, by averaging the k-corresponding scores of the batches associated with that chunk.
  Concatenate the previous step's results and return the signal $Sig(d)$ of the document d.
end for

3.2 Signals comparison and ranking

The ongoing research applies DTW distance to assess dissimilarity between signals. DTW distance has been

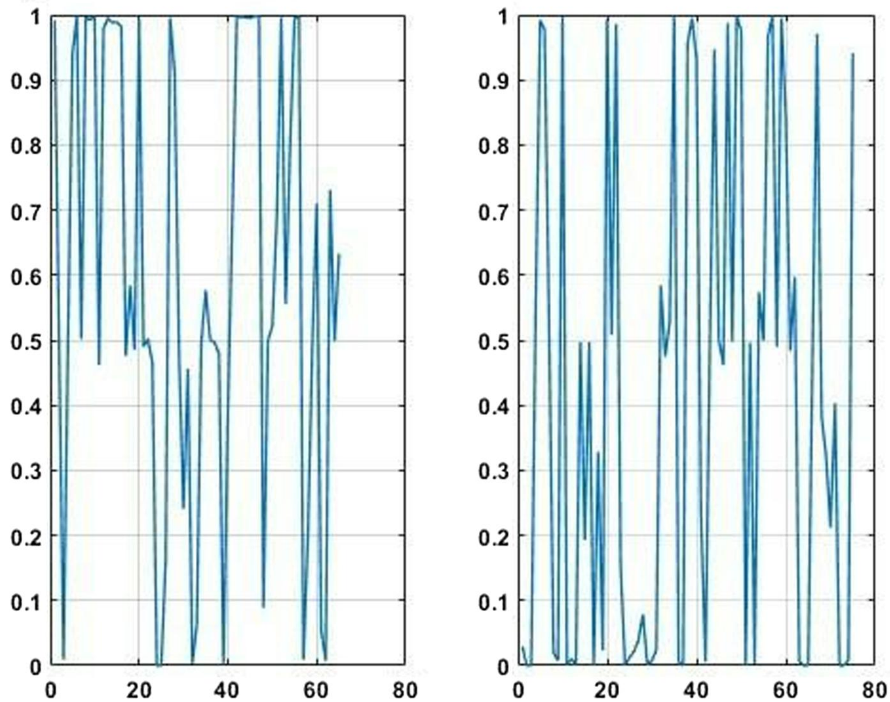

Figure 2. Examples of signals of 'The Tragedy of Romeo and Juliet' (the left panel) and 'The Tragedy of Hamlet Prince of Denmark' (the right panel) found for k = 2.

independently introduced by Sakoe and Chiba (1978) and Vintsyuk (1968). The *DTW* algorithm aims to find an optimal alignment between two temporal sequences, such as time series or signals, by aligning their time locations and minimizing the Euclidean distance within the aligned series. It utilizes 'elastic' transformations to identify comparable templates with different phases, making it a valuable method for assessing signal correspondence. For example, in Fig. 3, the calculated *DTW* distance value between the two signals is approximately 1.45.

We use a pairwise distance matrix constructed for all the documents investigated to identify anomalous documents. This matrix is generated using a Kuratowski-type embedding procedure (Kuratowski 1935), where each text is associated with a corresponding row in the distance matrix.

Anomaly detection in our approach handles the well-known Isolation Forest methodology initially introduced by Liu, Ting, and Zhou (2008). This unsupervised approach is designed to separate outliers from most data by employing a binary tree ensemble. Isolation Forest employs binary trees, called Isolation Trees (iTrees), where each node performs a binary split based on a randomly chosen feature and a randomly determined threshold. The algorithm constructs a set of such trees, randomly selecting attributes based on an arbitrarily chosen value within the range of that attribute's maximum and minimum values. A tree is constructed until all the documents in the collection have been examined or the number of splits exceeds the defined limit. The number of splits required to separate a point along the path from the tree root to the stopping leaf represents the anomaly score, which is averaged across all constructed trees.

3.3 Classifiers training

A separator *S* between two text corpora refers to a boundary or hyperplane in appropriate Euclidian space that distinguishes the elements of one set from those of another. Separators can be constructed using various machine learning techniques, such as *Support Vector Machines* (*SVMs*), logistic regression, neural networks, and *BERT* models. However, *SVMs* and logistic regression approaches are deemed infeasible in this study due to the complex configuration of text embedding in the corresponding space.

3.3.1 *CNN* based classifier

A neural network can undergo training using labeled data from both sets to construct a separator. Through an iterative learning process, a network adjusts its internal parameters (weights and biases) to minimize the prediction error and improve its ability to distinguish

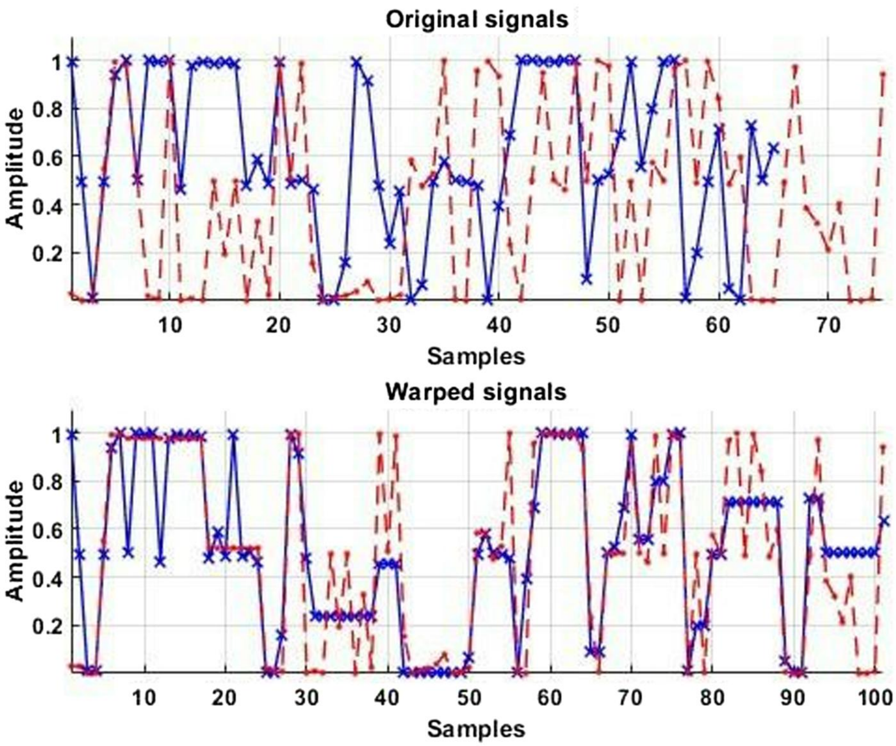

Figure 3. Example of *DWT* alignment of signals corresponding to 'The Tragedy of Romeo and Juliet' (the dark line) and 'The Tragedy of Hamlet Prince of Denmark' (the dotted line).

between the sets. The neural network's architecture can vary depending on the problem's complexity and the data's nature.

CNNs, traditionally utilized for image recognition and computer vision tasks, can also be adapted for textual data by employing appropriate embedding techniques. This adaptation enables to effectively process and analyze text data, leveraging their ability to capture hierarchical patterns and features.

To capture short patterns in text, the proposed methodology employs as one of the tools a *CNN* architecture inspired by prior research (see, e.g. Goldberg 2016; Golberg et al. 2017; Shrestha et al. 2017). *CNN* takes a sequence of matrices resulting from a word embedding as its input, enabling it to analyze and identify patterns within the text.

More precisely, after preprocessing and word embedding, the network employs a *1D* convolutional layer with three shift filters to capture short patterns. Then, a bi-directional *LSTM* is utilized to uncover sequential prototypes within the data. Finally, the network is completed by two dense layers that facilitate classifying the input into two distinct categories. A scheme of such a network is presented in Fig. A.1.

3.3.2 *BERT*-based classifier

A pre-trained *BERT* model can be used as an alternative to the *CNN*-based approach. As previously discussed, *BERT* is pre-trained on an extensive text corpus, enabling it to capture the nuances of language and comprehend the relationships between words. This approach proves to be beneficial for sentiment analysis tasks. During the fine-tuning process, inspired by the methodology introduced by Howard and Ruder (2018), a specialized layer is added on top of the pre-trained *BERT* model, designed explicitly for sentiment classification. The fine-tuned model is then trained on a dataset comprising both positive and negative text samples.

In our particular scenario, batches of texts extracted from one impostor group are labeled positive, while those from another are labeled negative. This labeling process, which mimics sentiment classification, is employed to fine-tune the underlying pre-trained *BERT* model. As a result of this adjustment, the *BERT* model is suggested to become skilled at analyzing new text data consisting of batches of the tested text collection.

3.4 Procedure summarizing

The procedure, applicable to both *CNN*-based and *BERT*-based approaches, generates a vector for each tested document containing its inner in-iteration anomaly scores. In this research, we apply a score offset strategy, aligning outliers with the decision function's negative values to effectively quantify each anomaly manuscript, indicating the 'likelihood' that it is not written in the expected style. It is important to emphasize that, based on our general perspective, most of the considered creations are deemed genuine, meaning they adhere to the underlying basic style. These authentic documents have to belong to the data core and exhibit predictable score patterns. On the contrary, the anomalous documents within the studied collection are anticipated to demonstrate similar characteristics within these scores but are distinguishable from those exhibited by the genuine texts.

Another perspective for interpreting the resulting score values is from a probability standpoint. Scores corresponding to authentic documents can be viewed as samples drawn from a population representing texts in the underlying style. Similarly, score values generated for deceptive texts can be seen as samples extracted from another population. This situation is framed as a two-sample problem, where the goal is to differentiate between the two populations. This gives rise to two distinct clusters, the genuine and the artificial ones, of creations, so the paired t-test statistic, calculated for two sets of scores, is used as a distance measure between the scores vectors.

Based on this understanding, the final score vectors are divided into clusters through the famous K-medoids algorithm in our application. Using a straightforward alignment within the iterative procedure, the cluster of documents suspected to be fake is labeled as 1, while the other cluster is labeled as 2.

4. Numerical experiments

The numerical experimental assessment uses a two-pronged approach. One part focused on evaluating the accuracy of the model. To assess the model's accuracy, we tested it on a dataset where the correct classifications are already provided (ground truth). We evaluated the model's capacity to uncover hidden structures by contrasting its predicted classifications with the known labels. The impostors' collections involved in the experiments are listed in Table A.1. As we will see, the system under examination demonstrates predictable behavior in both *CNN* and *BERT* varieties, consistently stabilizing as more impostor pairs are incorporated into the training process.

The approach utilizes materials incorporated from impostors, such as modern literary English texts, processed with methods like *Word2Vec* and *CNN* or a pre-trained transformer *BERT*. The paper operates on the premise that while language naturally evolves, the core stylistic elements and vocabulary, which form a significant part of the language's 'skeleton', tend to remain sufficiently resistant to changes. Thus, relatively modern textual material can be used in our methodology to analyze medieval manuscripts.

The source for all texts is Project Gutenberg, a well-established online archive. https://www.gutenberg.org/.

4.1 Model validation study

In this part, the following parameters are used in the training procedure of *CNN*:

- *Kernel sizes*—the kernel sizes in the *1D* convolution step are *3, 6, 12*,
- *Nb filter*—the number of filters in the *1D* convolution step is *500*,
- *Pool size*—the pooling size in the *1D* convolution step is *1*,
- *Num LSTM*—the number of units in the Bidirectional (*BI-LSTM*) step is *500*,
- *Dropout*—the dropout rate in the fully connected step is *0.25*,
- *Input fully*- connected layer—the number of cells in the input fully connected layer is *1024*,
- *Learning rate*—the procedure learning rate is *0.001*,
- Optimization method—*Adam*,
- Optimization parameters:
  - $momentum = 0.9$,
  - $decay = 1$,
- The number of epochs is *5*,
- The batch size, $L=50$,
- The number of batches in a chunk $k=8$.

In this validation study, we use impostor collections, presented in the Table 1.

Since target works are generally not large, all tested files are divided into approximately 80-kb parts to align with the future Shakespeare-focused analysis. Any remaining text portion is treated separately if it is sufficiently large.

4.1.1 John Galsworthy and Charles Dickens

The first tested collection includes:

1) John Galsworthy. The Forsyte Saga: Indian Summer of a Forsyte-104,282K,
2) John Galsworthy, The Forsyte Saga: Over the River-503,916K,

Table 1. The impostors' collection involved in the simulation study.

Number	Impostor
1	Charlotte Bronte
2	Benjamin Disraeli
3	Francis Bacon
4	Galsworthy
5	George Eliot
6	Harry Potter
7	Jane Austen
8	John Dryden
9	John Milton
10	Lord of the Rings
11	Taylor Swift
12	Thomas Hardy
13	Thomas More

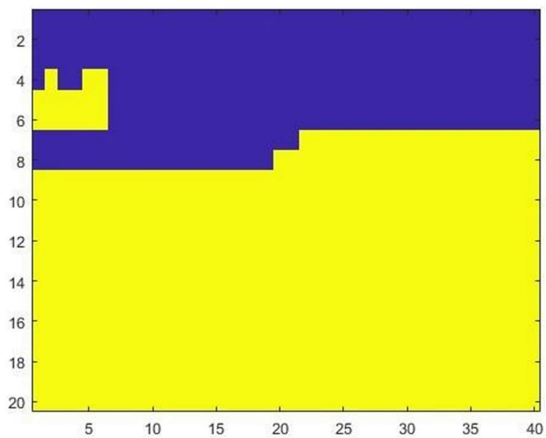

Figure 4. Evolution of cluster assignments in the first simulation study.

3) John Galsworthy, The Forsyte Saga: Flowering Wilderness-447,252K,
4) Charles Dickens, Children Stories-124,574K,
5) Charles Dickens, A Christmas Carol-189,067K.
6) text to classify.txt—external file included as sanity checks 123,740K.

The evolution of cluster assignments is given in Fig. 4.

This figure illustrates the clustering analysis outcomes performed across all sequential iterations. Label 1 (marked in dark) indicates works presumably not composed by John Galsworthy, while Label 2 (marked in bright) signifies works presumably composed by him. Table 2 presents the average assignment labels calculated for each creation throughout the procedure.

Texts with scores below 1.5 are highlighted in bold and identified as not being authored by Galsworthy. The distribution of works within the clusters

Table 2. Cluster averaged labels of the first collection.

Creation	Score
A Christmas Carol 2	**1**
Children Stories 0	**1**
text to classify	**1**
A Christmas Carol 1	**1.075**
Flowering Wilderness 5	**1.15**
Over the River 1	**1.15**
Children Stories 1	**1.475**
A Christmas Carol 0	1.525
Flowering Wilderness 0	2
Flowering Wilderness 1	2
Flowering Wilderness 2	2
Flowering Wilderness 3	2
Flowering Wilderness 4	2
Indian Summer 0	2
Indian Summer 1	2
Over the River 0	2
Over the River 2	2
Over the River 3	2
Over the River 4	2
Over the River 5	2

accurately reflects their original authorship. Specifically, two of the three sections of 'A Christmas Carol' are attributed to Charles Dickens. In contrast, only one of the six parts of 'Flowering Wilderness' is included in this category.

4.1.2 Francis Bacon and Christopher Marlowe

This artificial collection includes before parsing the following creations:

1) The Essays or Counsels, Civil and Moral, of Francis Ld. Verulam Viscount St. Albans by Francis Bacon-303,818K,
2) The Advancement of Learning by Francis Bacon-497,595K,
3) Valerius Terminus: of the Interpretation of Nature by Francis Bacon-86,065K,
4) Tamburlaine The Great-The Second Part, By Christopher Marlowe-131,563K,
5) text to classify—external file included as sanity checks 123,740K.

The assignment evolution is displayed as follows:

Table 3 exposes the mean assignment labels calculated for each creation throughout the procedure. The evolution of cluster assignments is given in Fig. 5.

Two distinct clusters are evident in the classification. Overall, this arrangement aligns with the underlying partition. However, it is intriguing to note the behavior of 'Essays' assigned to clusters that favor styles other than Bacon's. This may be attributed to the inception of Francis Bacon's literary journey in 1597 with the release of 'Essayes: Religious Meditations ... '.

Table 3. Cluster averaged labels of the second collection.

Creation	Score
text to classify	**1.073**
The Essays or Counsels, Civil and Moral 2	**1.073**
Tamburlaine the Great 1	**1.073**
Tamburlaine the Great 0	**1.073**
The Essays or Counsels, Civil and Moral 3	**1.073**
The Essays or Counsels, Civil and Moral 1	**1.146**
The Advancement of Learning 0	**1.317**
The Advancement of Learning 2	1.927
The Advancement of Learning 3	1.927
The Advancement of Learning 5	1.927
The Advancement of Learning 4	1.927
Valerius Terminus Of the Interpretation of Nature 0	1.927
The Advancement of Learning 1	1.927
The Essays or Counsels, Civil and Moral 0	1.927
The Advancement of Learning 6	2.000

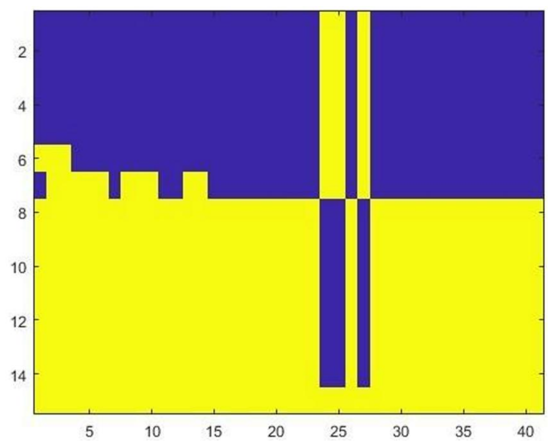

Figure 5. Evolution of cluster assignments in the second simulation study.

This initial collection, his debut publication and the foundation of a continually evolving project, showcased Bacon's remarkable versatility as a writer. His essays transcended stylistic boundaries, ranging from clarity and directness to wit and conciseness, often employing epigrams—clever and succinct expressions—to convey his ideas effectively. Over nearly three decades, from 1597 to 1625, Bacon continuously published and refined his 'Essays', ensuring their lasting influence. These essays span a spectrum of styles, from the straightforward and unadorned to the epigrammatic (see, e.g. Peltonen 1996).

4.2 Analysis of Shakespeare's collection

The examined collection consists of forty-nine literary works credited to William Shakespeare, while the impostor collection consists of thirty sets by different English writers, as given in Table A.1. As mentioned, all these texts are sourced from the reputable website Project Gutenberg (https://www.gutenberg.org/). The experimental study of this text collection is conducted using both a *CNN* architecture and a *BERT*-based approach.

One crucial parameter that significantly influences the results in both cases is the batch size, denoted as L, and the number of batches in a chunk, denoted by k. It is essential to settle between a batch size that is long enough to capture sufficient linguistic stability and a batch size that is short enough to maintain the length of the signal. Similar considerations are taken into account when determining the chunk span. Short chunks may result in a lack of diversity in the constructed signals, making it challenging to distinguish between them. Conversely, longer chunks produce shorter signals, which may decrease the process's steadfastness. By the parameters employed in the earlier simulation study, we establish $L = 50$ and $k = 8$ for the present investigation.

4.2.1 *CNN* classifier model

The parameters setup is following:

- *Kernel sizes*—the kernel sizes in the *1D* convolution step are *3, 6, 12*,
- *Nb filter*—the number of filters in the *1D* convolution step is *500*,
- *Pool size*—the pooling size in the *1D* convolution step is *1*,
- *Num LSTM*—the number of units in the Bidirectional (*BI-LSTM*) step is *500*,
- *Dropout*—the dropout rate in the fully connected step is *0.25*,
- *Input fully*—connected layer—the number of cells in the input fully connected layer is *2048*,
- *Learning rate*—the procedure learning rate is *0.001*,
- *Optimization method*—the network optimization method is *Adam*,
- *Optimization parameters* are momentum = *0.9*, and decay = *1*,
- *The number of epochs* is *5*.

In this case, the network's training procedure involves 41,993,038 trainable parameters. To ensure training process stability, chunks of the impostors' texts are duplicated, effectively doubling the volume for each impostor. It is a standard text augmentation technique.

The presented results are based on 133 iterations sampled from 435 possible distinct pairs of impostors. Like previously considered studies, the process stabilizes, allowing us to obtain results without losing generality by limiting ourselves to this sample from the overall population of impostor pairs. Not all classifiers meet the

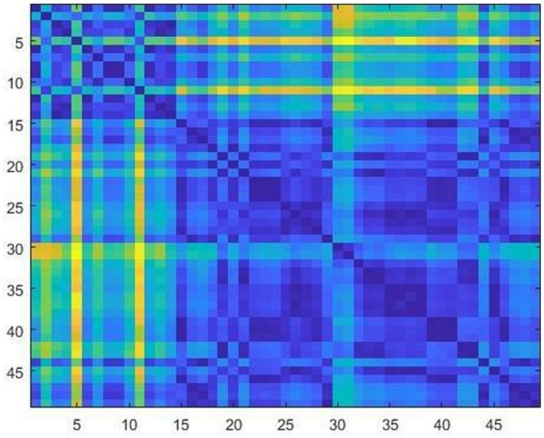

Figure 6. A matrix of pairwise distances arranged based on the assigned clusters of scores.

evaluation criteria for differing between impostors in a pair, assigning all creations just to one of the impostors. To ensure the integrity of our analysis, we removed such outcomes from consideration. As a result, the number of classifiers is reduced to 127, which aligns with the number of iterations in the approach.

Figure 6 demonstrates the grouping of the data points according to their cluster assignments. As usual, lighter tones refer to higher values.

The corresponding partition is given in Table 4. The first cluster identified is highlighted in bold.

To further understand and consider the varying importance of different characteristics and impostors' pairs in the classification process, we track the evolution of the clusters throughout the considered iterations. By observing how clusters change and stabilize over time, we can gain essential insights into understanding the global cluster assignments.

In Fig. 7, you can see the cluster assignments being evaluated at each step, along with the display of cluster coordination. The smaller cluster is Cluster 1, while the larger one is Cluster 2. Notably, the analysis begins from the 10th iteration.

Upon observation, it becomes evident that the process stabilizes quickly after a limited number of initial iterations. Figure 8 and Table 5 depict the averaged cluster labeling derived from the evolution presented in Fig. 7. Based on our analysis, an average cluster label of less than 1.5 indicates suspicious creations.

Within this context, it is significant to highlight the existence of two distinct groups. The first group consists of sixteen creations with scores below 1.5,

Table 4. Cluster assignment obtained based on all iterations.

Creation	Assignment
A LOVERS COMPLAINT	**1**
A MIDSUMMER NIGHT'S DREAM	**1**
A YORKSHIRE TRAGEDY BY SHAKESPEARE	**1**
ARDEN OF FEVERSHAM	**1**
KING HENRY V	**1**
KING HENRY VI Part III	**1**
KING JOHN	**1**
THE LIFE OF TIMON OF ATHENS	**1**
THE LONDON PRODIGAL	**1**
THE MERRY WIVES OF WINDSOR	**1**
THE PURITAINE WINDOW BY SHAKESPEARE	**1**
THE TRAGEDY OF ANTONY AND CLEOPATRA	**1**
THE WINTERS TALE	**1**
VENUS AND ADONIS	**1**
AS YOU LIKE IT	2
CYMBELINE	2
FAIR EM BY SHAKESPEARE	2
KING EDWARD III BY SHAKESPEARE	2
KING HENRY IV, PART 1	2
KING HENRY VI PART I	2
KING HENRY VI PART II	2
KING HENRY VIII	2
KING RICHARD II	2
KING RICHARD III	2
LOCRINE MUCEDORUS BY SHAKESPEARE	2
LOVE_S LABOUR_S LOST	2
MEASURE FOR MEASURE	2
MERCHANT OF VENICE	2
MUCH ADO ABOUT NOTHING	2
PERICLES PRINCE OF TYRE	2
SIR JOHN OLDCASTLE BY SHAKESPEARE	2
SIR THOMAS MORE BY SHAKESPEARE	2
THE COMEDY OF ERRORS	2
THE HISTORY OF TROILUS AND CRESSIDA	2
THE LIFE AND DEATH OF THE LORD CROMWELL BY SHAKESPEARE	2
THE MERRY DEVIL OF EDMONTON BY SHAKESPEARE	2
THE RAPE OF LUCRECE	2
THE TAMING OF THE SHREW	2
THE TEMPEST	2
THE TRAGEDY OF CORIOLANUS	2
THE TRAGEDY OF HAMLET PRINCE OF DENMARK	2
THE TRAGEDY OF JULIUS CAESAR	2
THE TRAGEDY OF KING LEAR	2
THE TRAGEDY OF MACBETH	2
THE TRAGEDY OF OTHELLO MOOR OF VENICE	2
THE TRAGEDY OF ROMEO AND JULIET	2
THE TRAGEDY OF TITUS ANDRONICUS	2
THE TWO GENTLEMEN OF VERONA	2
THE TWO NOBLE KINSMEN BY SHAKESPEARE	2

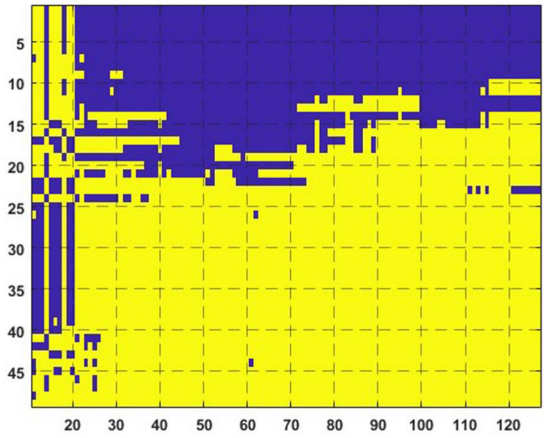

Figure 7. Evolution of cluster assignments.

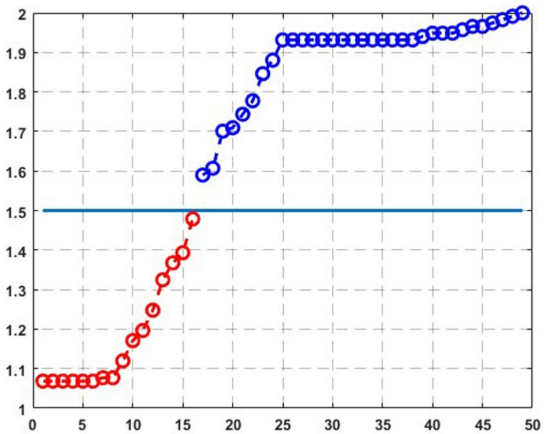

Figure 8. The averaged cluster labeling obtained using all iterations.

strongly suggesting they are attributed to William Shakespeare but are likely not composed by his distinctive style.

An additional evaluation of the obtained results can be conducted in line with the methodology outlined in Algorithm 1, proposed by Koppel and Winter (2014), as mentioned earlier. This approach offers a robust framework for further analysis, ensuring a comprehensive understanding of the data. Following this attendance, we introduce.

In data analysis, cluster labels denote the categories or identifiers assigned to groups of data points generated by clustering algorithms. Aligning cluster labels ensures that clusters derived from different clustering results correspond to similar data points. In our scenario, data are segmented into two clusters, and alignment may involve matching the cluster sizes, indicating small and large clusters correspondingly.

Algorithm 3 Authorship Verification Algorithm with randomly chosen impostor pairs

1: **Input:**
- *A*: A matrix $n \times N$ of document scores, where n is the number of the tested documents, and *N* is the number of the performed iterations,
- *T*: A number of steps in the procedure,
- *m:* A number of randomly chosen impostor pairs involved in each iteration.

2: **Procedure**
- Repeat *T* times:
  - Randomly choose *m* features without substitution from *N* collection features,
  - Clustering *n* documents into 2 clusters resting upon the chosen *m* features.
- Alignment of cluster labels,
- Calculate the average labels for each tested document across all performed iterations.

In our experiments, we use $m = 50$ and $T = 500$. Table 6 displays the results. A comparison of the results obtained by the three summarization methods is presented in Table 7.

As observed, all three methods consistently identify suspected fifteen creations. The creation 'King Henry VI, Part II' is detected only by the first method, while 'The Tragedy of Antony and Cleopatra' and 'The Winter's Tale' are detected by the second and third methods, respectively.

4.2.1.1 Several results and discussion

- '**The Comedy of Errors**' and '**The Merry Wives of Windsor**'

There are varying opinions regarding the authorship of the texts under examination. For instance, '**The Merry Wives of Windsor**', '**The Comedy of Errors**', and '**The Tragedy of Julius Caesar**' were identified as outliers in a study by Moshe Koppel and Shachar Seidman, which aimed to automatically identify pseudo-epigraphic texts (Kestemont et al. 2016). Our research corroborates this classification for '**The Comedy of Errors**' and '**The Merry Wives of Windsor**'. It should be noted that '**The**

Table 5. Partition of the creations based on the average cluster labels.

Creation	Score
A LOVERS COMPLAINT	**1.07**
ARDEN OF FEVERSHAM	**1.07**
KING JOHN	**1.07**
THE MERRY WIVES OF WINDSOR	**1.07**
THE PURITAINE WIDDOW BY SHAKESPEARE	**1.07**
VENUS AND ADONIS	**1.07**
A MIDSUMMER NIGHT_S DREAM	**1.08**
THE LONDON PRODIGAL	**1.08**
KING HENRY V	**1.12**
A YORKSHIRE TRAGEDY BY SHAKESPEARE	**1.17**
LOCRINE MUCEDORUS BY SHAKESPEARE	**1.20**
THE LIFE OF TIMON OF ATHENS	**1.25**
KING HENRY VI PART III	**1.32**
KING HENRY VI PART I	**1.37**
THE COMEDY OF ERRORS	**1.39**
KING HENRY VI PART II	**1.48**
THE RAPE OF LUCRECE	1.59
SIR JOHN OLDCASTLE BY SHAKESPEARE	1.61
THE TRAGEDY OF HAMLET PRINCE OF DENMARK	1.70
THE TRAGEDY OF JULIUS CAESAR	1.71
KING HENRY IV, PART 1	1.74
KING RICHARD III	1.78
THE WINTER’S TALE	1.85
AS YOU LIKE IT	1.88
CYMBELINE	1.93
KING EDWARD III BY SHAKESPEARE	1.93
KING HENRY VIII	1.93
KING RICHARD II	1.93
MEASURE FOR MEASURE	1.93
MUCH ADO ABOUT NOTHING	1.93
PERICLES PRINCE OF TYRE	1.93
THE HISTORY OF TROILUS AND CRESSIDA	1.93
THE TRAGEDY OF ANTONY AND CLEOPATRA	1.93
THE TRAGEDY OF CORIOLANUS	1.93
THE TRAGEDY OF KING LEAR	1.93
THE TRAGEDY OF OTHELLO MOOR OF VENICE	1.93
THE TWO GENTLEMEN OF VERONA	1.93
THE TWO NOBLE KINSMEN BY SHAKESPEARE	1.93
THE TRAGEDY OF ROMEO AND JULIET	1.94
FAIR EM BY SHAKESPEARE	1.95
THE TEMPEST	1.95
THE TRAGEDY OF MACBETH	1.95
LOVE_S LABOR’S LOST	1.96
MERCHANT OF VENICE	1.97
THE TRAGEDY OF TITUS ANDRONICUS	1.97
THE LIFE AND DEATH OF THE LORD CROMWELL BY SHAKESPEARE	1.97
THE MERRY DEVIL OF EDMONTON BY SHAKESPEARE	1.98
SIR THOMAS MORE BY SHAKESPEARE	1.99
THE TAMING OF THE SHREW	2.00

Table 6. Partition of the creations based on the average cluster sampled labels.

Creation	Score
THE MERRY WIVES OF WINDSOR	**1.29**
A LOVERS COMPLAINT	**1.29**
ARDEN OF FEVERSHAM	**1.29**
KING JOHN	**1.29**
THE PURITAINE WIDDOW BY SHAKESPEARE	**1.29**
VENUS AND ADONIS	**1.29**
KING HENRY V	**1.30**
THE LONDON PRODIGAL	**1.30**
THE LIFE OF TIMON OF ATHENS	**1.31**
KING HENRY VI PART III	**1.32**
A MIDSUMMER NIGHT_S DREAM	**1.34**
THE WINTERS TALE	**1.34**
LOCRINE MUCEDORUS BY SHAKESPEARE	**1.42**
THE TRAGEDY OF ANTONY AND CLEOPATRA	**1.43**
A YORKSHIRE TRAGEDY BY SHAKESPEARE	**1.44**
THE COMEDY OF ERRORS	**1.45**
KING HENRY VI PART I	**1.49**
KING HENRY VIII	1.51
THE TRAGEDY OF JULIUS CAESAR	1.53
THE RAPE OF LUCRECE	1.54
MUCH ADO ABOUT NOTHING	1.66
SIR JOHN OLDCASTLE BY SHAKESPEARE	1.67
THE TRAGEDY OF TITUS ANDRONICUS	1.68
THE TRAGEDY OF HAMLET PRINCE OF DENMARK	1.69
THE TRAGEDY OF CORIOLANUS	1.69
MERCHANT OF VENICE	1.70
AS YOU LIKE IT	1.70
CYMBELINE	1.71
LOVE_S LABOUR_S LOST	1.71
MEASURE FOR MEASURE	1.71
PERICLES PRINCE OF TYRE	1.71
THE TAMING OF THE SHREW	1.71
THE HISTORY OF TROILUS AND CRESSIDA	1.71
THE TEMPEST	1.71
THE TRAGEDY OF KING LEAR	1.71
THE TRAGEDY OF ROMEO AND JULIET	1.71
THE TWO GENTLEMEN OF VERONA	1.71
FAIR EM BY SHAKESPEARE	1.71
THE TRAGEDY OF OTHELLO MOOR OF VENICE	1.71
THE LIFE AND DEATH OF THE LORD CROMWELL BY SHAKESPEARE	1.72
THE TWO NOBLE KINSMEN BY SHAKESPEARE	1.72
KING EDWARD III BY SHAKESPEARE	1.73
KING RICHARD II	1.74
THE TRAGEDY OF MACBETH	1.74
KING HENRY VI PART II	1.74
THE MERRY DEVIL OF EDMONTON BY SHAKESPEARE	1.74
KING HENRY IV, PART 1	1.74
KING RICHARD III	1.75
SIR THOMAS MORE BY SHAKESPEARE	1.77

Table 7. Comparison of results obtained by three evaluation methods.

Creation	1	2	3
A LOVERS COMPLAINT	1	1	1
A MIDSUMMER NIGHT'S DREAM	1	1	1
A YORKSHIRE TRAGEDY BY SHAKESPEARE	1	1	1
ARDEN OF FEVERSHAM	1	1	1
KING HENRY V	1	1	1
KING HENRY VI PART I	1	1	1
KING HENRY VI PART III	1	1	1
KING JOHN	1	1	1
LOCRINE MUCEDORUS BY SHAKESPEARE	1	1	1
THE COMEDY OF ERRORS	1	1	1
THE LIFE OF TIMON OF ATHENS	1	1	1
THE LONDON PRODIGAL	1	1	1
THE MERRY WIVES OF WINDSOR	1	1	1
THE PURITAINE WINDOW BY SHAKESPEARE	1	1	1
VENUS AND ADONIS	1	1	1
KING HENRY VI PART II	1	0	0
THE TRAGEDY OF ANTONY AND CLEOPATRA	0	1	1
THE WINTERS TALE	0	1	1

[1] The "0" mark indicates that the text is not recognized as authentic, while the "1" mark signifies it was.

Tragedy of Julius Caesar' receives scores that are closely aligned with those of the pseudo-Shakespearean works depicted.

- '***Arden of Feversham***' has been a subject of intense scholarly debate regarding its authorship. Despite being published anonymously, various attributions and speculations concerning the author have emerged. Some sources, including the 2016 edition of The Oxford Shakespeare (referenced in 'Christopher Marlowe credited as one of Shakespeare's co-writers' at *theguardian.com*, October 23, 2016. The cite[1] (visited in 30.06. 2024)) associate '***Arden of Feversham***' with William Shakespeare and an anonymous collaborator, dismissing the possibility of Thomas Kyd or Christopher Marlowe being the sole authors. The play's inclusion in the Shakespeare Apocrypha indicates that some scholars consider it potentially linked to Shakespeare, even though the exact degree of his direct involvement remains uncertain. Our research supports this interpretation.
- '***The London Prodigal***' is considered a pseudonymous work, meaning it is attributed to William Shakespeare in some sources, but the authorship remains uncertain or disputed. This play is one of the works that fall into the category of Shakespeare's Apocrypha, which includes plays attributed to Shakespeare but not definitively confirmed as his creations. '***The London Prodigal***' was first published in 1605, and it is a comedy that deals with themes of prodigality and redemption. While it has been linked to Shakespeare in the past, most modern scholars do not consider it a genuine work of Shakespeare, and its authorship remains anonymous or uncertain. Therefore, it is classified as pseudo-Shakespeare in many literary and scholarly discussions.
- '***The life of Timon of Athens***' Since the 19th century, scholars have debated the authorship of 'Timon of Athens'. The play's unusual features have led some to believe it was written by two playwrights with distinct styles. The most popular candidate for co-authorship is Thomas Middleton, first suggested by John Jowett in Wells et al. (1997). Evidence advocates Shakespeare and Middleton collaborated on 'Timon' around 1605–1606. Eilidh Kane's article (Kane 2016) in the Journal of Early Modern Studies supports this timeframe. A morphological analysis by Taylor (1987) suggests a slightly later date for the Shakespearean portions, placing them between '**All's Well That Ends Well**' (1604–1605) and '**Macbeth**' (1606). In conclusion, the play's style deviates from Shakespeare's typical work, suggesting potential co-authorship. Generally, the lack of Shakespeare's characteristic style in '***The life of Timon of Athens***' strengthens the case for co-authorship, albeit indirectly.
- '***King Henry VI***'. As was mentioned in[2] 'Oxford University Press drew attention last year for deciding that, in the New Oxford Shakespeare, the plays Henry VI, Parts 1, 2, and 3 would no longer be listed as having been written by Shakespeare alone. Instead the title pages will say: "*By William Shakespeare and Christopher Marlowe*"'. This viewpoint is also supported in particular in[3] The New Oxford Shakespeare edition of the playwright's works published by[4] Oxford University Press online ahead of a worldwide print release lists also Christopher Marlowe as Shakespeare's co-author on the three 'Henry VI' plays, parts 1, 2, and 3. While this conclusion aligns with our research, further verification against established academic sources remains necessary. Notably, this study provides confirmation of co-authorship for Parts 1 and 3 through the application of all analytical methods. Co-authorship in Part 2 is established solely through the first method.
- '***Venus and Adonis***' The publication of '***Venus and Adonis***' in 1593 likely marks the beginning of William Shakespeare's literary career (see, e.g.

Kolin 1997) This debuts work, recognized as an anomaly in its distinct style by our research, may offer a glimpse into the evolution of Shakespeare's writing as he matured as a playwright. This play is also the subject of many considerations and studies. For instance, such research is presented in Stritmatter (2004). This article examines the history of scholarship on '*Venus and Adonis*', highlighting contradictions that question the traditional attribution to William Shakespeare. It argues that orthodox critics have failed to contextualize the poem psychologically and culturally properly. The alternative theory suggests Edward de Vere as the factual author but also presents interpretative challenges. The article explores the poem's dedication, cultural and political context, language, and imitations, ultimately providing a compelling rationale for the possible concealment of the actual author's identity.

- Project Gutenberg is a non-profit digital library offering over 60,000 free eBooks. It primarily focuses on older literary works whose copyrights have expired in the United States. According to the Gutenberg resource, works marked as 'by Shakespeare' are considered suspicious. Three of these plays fall into the suspended zone, while seven are recognized as genuine. It is important to note that the notable Project Gutenberg categorizes works based on the information available when they enter the public domain. Since Shakespeare's compositions have long been public, Project Gutenberg does not make definitive authorship claims reflecting existing attributions.

4.3 *BERT*-based classifier model

The *BERT* general framework (e.g. Devlin et al. 2018) offers another way to create a series of impostor base classifiers. In analogy with *CNN*'s presented applications, applying a *BERT*-based approach to tweet sentiment analysis in our study appears natural. Post-training in the context of Authorship Verification uses various approaches, one of the most popular of which is the Siamese architecture. This method, as demonstrated by Almutairi, Kang, and Al Hashimy (2023), involves a specific neural network design to compare two pieces of text to determine if they likely share the same author. A Siamese network consists of two identical sub-networks, often called 'twin networks', which can be any neural network architecture suitable for text processing, though *CNN*s or *BERT* models commonly used for *AV* tasks. For a comprehensive review of such methods, refer to Stamatatos et al. (2023). However, applying this method directly in our context poses challenges. These approaches typically require training with large amounts of data, which is not feasible in our case.

As previously mentioned, our study in this section is grounded in the *BERT*-based model, which has shown significant promise in various *NLP* tasks. For detailed implementation, we referred to the TensorFlow tutorial on text classification using *BERT*, available at

- https://www.tensorflow.org/text/tutorials/classify_text_with_bert.
- TensorFlow Text: Classify Text with *BERT* https://www.tensorflow.org/text/tutorials/classify_text_with_bert.

These tutorials provide a comprehensive guide on leveraging *BERT* for text classification, which we adapted and extended to suit the specific requirements of our research. A classification layer is the final step added on top of the pre-trained *BERT* model in a *BERT*-based approach for tasks like text classification. It takes the output generated by *BERT* and transforms it into a prediction about the text's category (an impostor in our case). This is a simple neural network with one hidden (of 512 neurons) and one classification layer (0,1). The architecture is given in Fig. A.2. Here is a step-by-step breakdown of the *BERT*-based approach given in Algorithm 4.

We also want to emphasize that the procedure presented here differs from the previously described method only in terms of signal construction. The results obtained are given in Table 8.

Thus, only eight initially suspected creations remain; see Table 9.

Moreover, a *BERT*-based classification has exposed several works commonly believed to be genuine as inauthentic. So, there appears to be a notable discrepancy between these results and the facts discussed earlier. This difference is apparently due to the fundamentally distinct training methodologies used for the two models in our two-fold analysis. The *CNN* model is built from scratch, relying solely on its general architecture's capabilities when trained with 'impostor materials' (data representing various writing styles). In contrast, the *BERT*-based model benefits from pre-training, inheriting substantial knowledge from a pre-trained configuration, thus starting with a solid foundation. A specifically trained classification layer is added to this pre-trained base to handle the task. It makes an impression that a more intensive post-training procedure is required to address this problem.

5. Conclusion

In conclusion, this work introduces a novel methodology for investigating the stylistic fingerprints of authorship. The approach goes beyond analyzing isolated words,

Algorithm 4 Authorship Verification Algorithm using a *BERT* model

Input:

D: texts under consideration.

S: A set of impostors.

Output: Authorship of text in *D*.

1. Load a pre-trained *BERT* model. In our research, it is https://tfhub.dev/tensorflow/small_bert/bert_en_uncased_L-10_H-768_A-12/1
2. Add a classification layer on top of the *BERT* model,
3. Iterate within impostors' parts *(i ,j):*
   a. Divide all texts under consideration $d \in D$ and impostors texts $z \in (Z_i, Z_j)$ into sequential batches containing *L* terms,
   b. Label batches for training as positive for Z_i, and negative for Z_j,
   c. Fine-tune the pre-trained *BERT* model on the labeled batches to obtain a separator $S_{i,j}$. by training the added classification layer using labeled impostors' batches,
   d. Distinctly classifies the bathes of $d \in D$ to different impostor collections by appropriate labeling,
   e. Construct a signal *S(d) for each* $d \in D$
   f. Classify all signals *as described in Algorithm 1*.
4. Classify all documents $d \in D$ according to results obtained at each iteration.

encompassing intricate patterns across multiple linguistic structures. We establish innovative projection frameworks by constructing and applying prototypes derived from classifiers trained on impostor text pairs. These outlines enable the differentiation of complex stylistic nuances within language. The case study of the Shakespearean authorship question compellingly demonstrates this methodology's efficacy in resolving longstanding literary debates. Notably, within a *CNN* paradigm, the system can discern pseudo-Shakespearean creations, whether falsely attributed or jointly authored. So far, applying *BERT*-based approaches has resulted in less satisfactory outcomes. We suggest this is due to an insufficiently effective post-training procedure. In future studies, we aim to address this issue by employing multilayer post-training layers and methods similar to *BiBERT-AV*.

Shakespearean scholarship advances through a dynamic exchange of ideas. The diverse perspectives and extensive expertise of scholars in this field play a crucial role in uncovering new interpretations, questioning

Table 8. Partition of the creations based on the average cluster labels using a *BERT*-based architecture.

Creation	Score
A LOVERS COMPLAINT	**1.28**
LOVE_S LABOUR_S LOST	**1.28**
THE HISTORY OF TROILUS AND CRESSIDA	**1.28**
THE MERRY DEVILL OF EDMONTON BY SHAKESPEARE	**1.29**
A MIDSUMMER NIGHT'S DREAM	**1.30**
ARDEN OF FEVERSHAM	**1.33**
KING HENRY VIII	**1.33**
THE RAPE OF LUCRECE	**1.33**
VENUS AND ADONIS	**1.34**
KING HENRY VI PART I.	**1.35**
LOCRINE MUCEDORUS BY SHAKESPEARE	**1.36**
THE WINTERS TALE	**1.36**
THE TRAGEDY OF MACBETH	**1.43**
THE LIFE OF TIMON OF ATHENS	**1.47**
A YORKSHIRE TRAGEDY BY SHAKESPEARE	**1.48**
THE TRAGEDY OF HAMLET PRINCE OF DENMARK	1.61
KING HENRY VI PART III	1.68
KING HENRY VI PART II	1.70
MUCH ADO ABOUT NOTHING	1.71
MEASURE FOR MEASURE	1.72
KING HENRY V	1.72
CYMBELINE	1.72
KING HENRY IV, PART 1	1.72
KING EDWARD III BY SHAKESPEARE	1.72
KING JOHN	1.72
KING RICHARD II	1.72
PERICLES PRINCE OF TYRE	1.72
SIR THOMAS MORE BY SHAKESPEARE	1.72
THE LIFE AND DEATH OF THE LORD CROMWELL BY SHAKESPEARE	1.72
THE TAMING OF THE SHREW	1.72
THE TRAGEDY OF KING LEAR	1.72
THE TRAGEDY OF OTHELLO MOOR OF VENICE	1.72
THE TRAGEDY OF ROMEO AND JULIET	1.72
THE TRAGEDY OF TITUS ANDRONICUS	1.72
THE MERRY WIVES OF WINDSOR	1.73
THE COMEDY OF ERRORS	1.73
THE TRAGEDY OF ANTONY AND CLEOPATRA	1.78
THE TRAGEDY OF JULIUS CAESAR	1.79
AS YOU LIKE IT	1.84
THE PURITAINE WIDDOW BY SHAKESPEARE	1.85
FAIR EM BY SHAKESPEARE	1.86
KING RICHARD III	1.89
THE TEMPEST	1.89
SIR JOHN OLDCASTLE BY SHAKESPEARE	1.92
THE TWO GENTLEMEN OF VERONA	1.93
THE LONDON PRODIGAL	1.94
THE TRAGEDY OF CORIOLANUS	1.94
THE TWO NOBLE KINSMEN BY SHAKESPEARE	1.94
MERCHANT OF VENICE	1.97

Table 9. Creations recognized as suspicious by the *CNN* and *BERT*-based methods.

Creation
A LOVERS COMPLAINT
A MIDSUMMER NIGHT'S DREAM
ARDEN OF FEVERSHAM
VENUS AND ADONIS
KING HENRY VI PART I
LOCRINE MUCEDORUS BY SHAKESPEARE
THE LIFE OF TIMON OF ATHENS
A YORKSHIRE TRAGEDY BY SHAKESPEARE

existing assumptions, and enhancing our comprehension of William Shakespeare's enduring influence. We are keen to obtain the insights of these scholars on the findings presented in our research. This collaborative effort, where rigorous academic investigation intersects with the knowledge of established experts, is likely to produce the most profound and nuanced exploration of Shakespeare's work. We would greatly value their assessments of the results obtained in our research.

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Zeev Volkovich (Conceptualization, Formal Analysis, Software) and Renata Avros (Formal Analysis, Software, Validation)

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Notes

1. https://www.theguardian.com/culture/2016/oct/23/christopher-marlowe-credited-as-one-of-shakespeares-co-writers.
2. https://www.folger.edu/podcasts/shakespeare-unlimited/christopher-marlowe-attribution-henry-vi/.
3. https://www.smithsonianmag.com/smart-news/what-know-shakespeares-newly-credited-collaborator-chistopher-marlowe-180960902/?utm_source=chatgpt.com.
4. https://global.oup.com/?cc=us.

Appendix

Models architectures are presented in the Figs A.1 and A.2.

Table A.1 presents the authors whose creations are used as impostors.

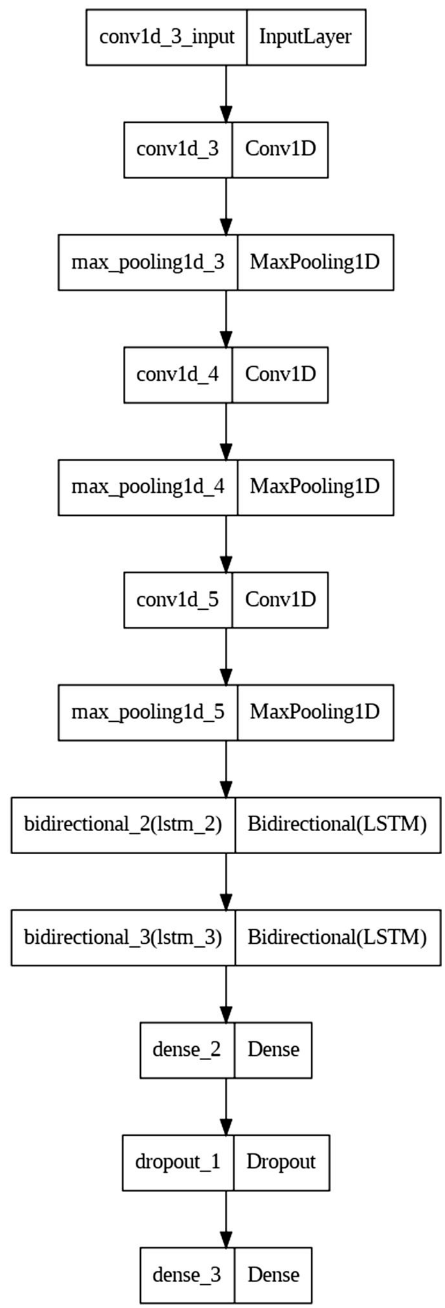

Figure A.1. The architecture of the applied *CNN* Network.

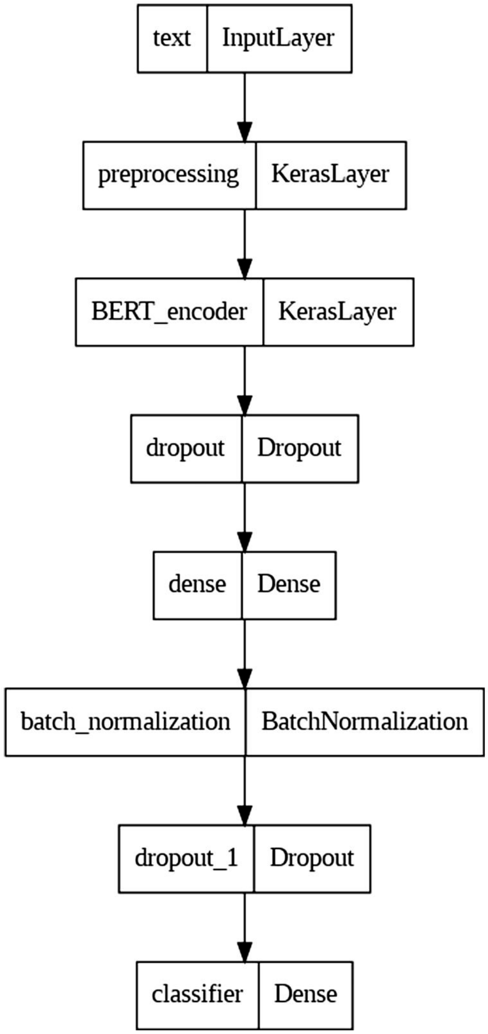

Figure A.2. The architecture of the *BERT* post-training procedure.

Table A.1. List of impostors.

N	Creation
1	Agatha Christie
2	Arthur C Clarke
3	Arthur Ignatius Conan Doyle
4	Benjamin Disraeli
5	Benjamin Jonson
6	Charles Dickens
7	Charlotte Brontë
8	Christopher Marlowe
9	Elizabeth Cleghorn Gaskell
10	Fanny Burney
11	Francis Bacon
12	Galsworthy
13	Geoffrey Chaucer
14	George Eliot
15	Harry Potter

(continued)

Table A.1. (continued)

N	Creation
16	Isaac Asimov
17	Jane Austen
18	John Donne
19	John Dryden
20	John Milton
21	Lewis Carroll
22	Lord of the Rings
23	Mark Twain
24	Mary Shelley
25	Robert Sheckley
26	Taylor Swift
27	Thomas Hardy
28	Thomas Kyd2
29	Thomas More
30	Virginia Woolf

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